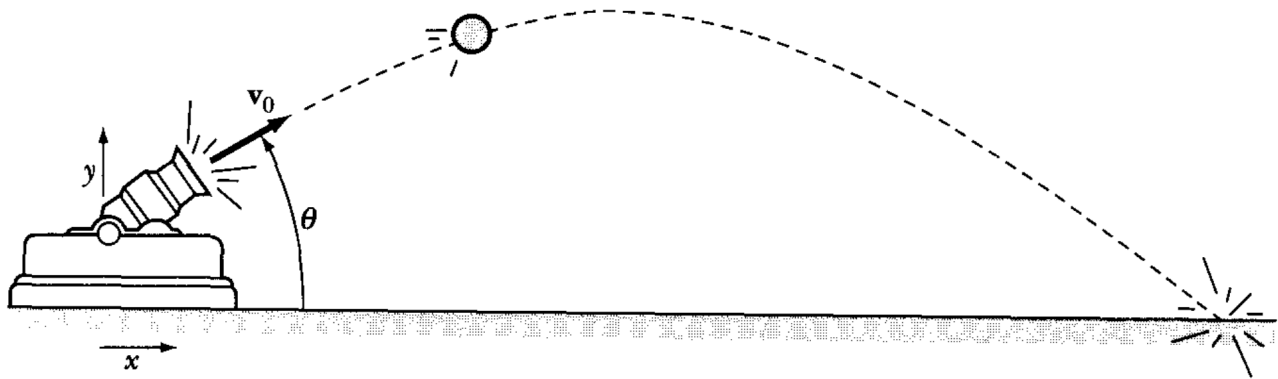


**Advanced Classical Mechanics (PHYS 615)**  
**Extra Credit Project 1 (due on date of Exam I)**

**Fall 2026**

The points obtained in this project will be added towards Exam I. To obtain full credit, your computer source code + calculations done on paper **MUST** be provided along with the plots.

Projectile Motion with Drag: **(5 pts)** Consider a projectile motion with drag force of the form  $\mathbf{F} = -km\mathbf{v}$



Show that the projectile flight time and range are

$$T = \frac{kV + g}{gk}(1 - e^{-kT}) \quad (1a)$$

$$R = \frac{U}{k}(1 - e^{-kT}) \quad (1b)$$

where  $k$  is a drag coefficient and  $V \equiv v_0 \sin \theta$  and  $U \equiv v_0 \cos \theta$  are the initial y- and x-components of the projectile speed.

Note that the time of flight  $T$  is a transcendental equation and can't be solved analytically. Two methods for solving this problem are (i) *Perturbative* and (ii) *Numerical*

Perturbative Approach: (5 pts)

To use the perturbation method, we find an expansion parameter coupling constant that is normally small. In the present case, this parameter is the retarding force constant  $k$ , because we have already solved the present problem with  $k = 0$  in the lecture, and now we would like to turn on the retarding force, but let  $k$  be small. We therefore expand the exponential term  $e^{-kT}$  in the flight time  $T$  (Eq. 1a) in a power series with the intention of keeping only the lowest terms of  $k^n$ , where  $k$  is our expansion parameter. Keep only terms in the expansion through  $k^3$  and show it can be expressed as

$$T = \frac{2V/g}{1 + kV/g} + \frac{1}{3}kT^2 \quad (2)$$

Note that we now have the expansion parameter in the denominator of the first term, which we need to expand this term in a power series  $1 / (1 + kV/g) \sim \dots$  keeping only terms through  $k^2$ , since we only have terms through  $k$  in the  $T$  expression above. Now substitute the power series and keep only terms in  $k$  to first order to show that the flight time simplifies to

$$T = \frac{2V}{g} + \left( \frac{T^2}{3} - \frac{2V^2}{g^2} \right) k + O(k^2) \quad (3)$$

where we choose to neglect  $O(k^2)$ , the terms of  $k^2$  and higher order.

Note from this expression, in the limit where  $k \rightarrow 0$ , the flight time reduces to the expected expression  $T_0 = 2V/g$  for a projectile without air drag.

If  $k$  is small enough (but not vanishing), we can approximate  $T \approx T_0$ , which upon substituting in the simplified flight time (Eq. 2), show that it yields the approximate flight time

$$T \approx \frac{2V}{g} \left( 1 - \frac{kV}{3g} \right) \quad (4)$$

Now to obtain an approximate expression for the range, apply a Taylor expansion to the term  $e^{-kT}$  in the range (Eq.1b) using  $k$  as the *expansion parameter* and keeping terms only up  $k^2$ , to show it yields the expression

$$R \approx \frac{U}{k} \left( kT - \frac{1}{2}k^2T^2 + \frac{1}{6}k^3T^3 - \dots \right) \quad (5)$$

Now substitute the approximate expression for the flight time (Eq. 4) to the range (Eq. 5) keeping only up to  $k^2$  terms, and re-arrange the equation to show that

$$R \approx R_0 \left[ 1 - 4\left(\frac{kV}{3g}\right) + 9\left(\frac{kV}{3g}\right)^2 \right] \quad (6)$$

where  $R_0 \equiv 2UV/g$  (projectile range without drag)

Numerical Approach: (10 pts) To solve a transcendental equation, we use *Newton's Method* numerical approach which is a root-finding algorithm, which has the recurring form:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where  $x_{n+1}$  is the root after  $n$ -recursive steps. To use this method, the function must be of the form  $f(x) = 0$  and it requires an initial guess of the root  $x$  that satisfies the equation. In our case, for a given value of  $k$ , we find the root  $T$ , and used the  $(k, T)$  pair as input into the range equation, (Eq. 1b) .

Solve the transcendental equation using Newton's Method.

**step 1:** need to put (Eq. 1a) in the form:  $f(k, T) = 0$

**step 2:** for an arbitrary value of  $0 < k \leq 0.08$ , find the root  $T$ , as follows

$$T_{n+1} = T_n - \frac{f(k, T_n)}{df(k, T_n)/dT}$$

Use the initial conditions to get the iteration process going, and try using  $n=100$  iterations

initial conditions ( $n=0$ ):

$$v_0 = 600 \text{ m/s}$$

$$\theta = 60^\circ$$

$$T_0 = 150 \text{ sec (initial guess will not necessarily satisfy } f(k, T_0) = 0)$$

after  $n$ -iterations, the algorithm should have found the root that satisfies the equation  $f(k, T) = 0$ ,

**step 3:** now that you have a  $(k_i, T_i)$  pair for an  $i$ -th arbitrary value of  $k$ , use it as input into the range equation,  $R_i(k_i, T_i)$

**step 4:** at this point, you should have a procedure established for determining the values  $(k_i, T_i, R_i)$ ; now repeat the procedure for a range of  $k$ -values from 0.0000001 to 0.08 with step-size of  $dk = 0.001$ ; \*\* note: since  $k$  is in the denominator,  $k \neq 0$

**step 5:** make a plot of  $R$  vs.  $k$  to examine how the projectile range varies with the drag coefficient \*\* note: negative ranges are unphysical, therefore, when making the plots, set the lower limit on the y-axis to zero.  
Add proper labels and a legend to each of the plots.

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Now, to check the *perturbative* versus the *numerical* method, make a plot overlay of:

- i)  $R$  vs.  $k$  (*perturbative* **linear** approximation) [keep only linear term in (6)]
- ii)  $R$  vs.  $k$  (*perturbative* **quadratic** approximation) [linear term + quadratic]
- iii)  $R$  vs.  $k$  (**numerical** approximation)

Add proper labels and a legend to each of the plots.

Write a brief paragraph commenting on your results. Is it what you expected to happen ? Elaborate.